

PALW v0.6.1: Peer-to-Peer Proof of Useful Work from Large-Language-Model Computation

Testnet-23 Safety Profile: Producing a One-Time MatureTicket from Canonical LLM Computation

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Abstract

Converting costly LLM computation into a reward-bearing object in a peer-to-peer network without a trusted coordinator requires the computation to be fixed before the lottery, to yield one-time trials proportional to the committed work, and to expose evidence that independent nodes evaluate identically. This paper defines PALW v0.6.1, a protocol that converts canonical LLM execution into a one-time MatureTicket.

A worker groups several executions of one Compute Set and one shape into a homogeneous BatchCommit. Every execution separates a result salt that hides the projection from a lottery-only salt commitment. After the BatchCommit, a future window of ALGO-3 blocks on the GHOSTDAG selected chain determines lottery randomness, and only a winner publishes a WinAnnounce. The WinAnnounce reveals the lottery salt, one winning trial, the selected job bytes, a Merkle path, and the frozen target reference. Only the first valid announcement for a BatchCommit is accepted in branch-local state. Verification randomness and the replica roster do not depend on the winning index, so choosing among multiple winning trials does not grind audit or replica draws.

Authenticity is checked in three layers. Layer 0, the Non-Interactive Sampled Audit (NISA), is a validity rule under which full nodes re-execute intervals selected by future randomness. Layer 1 treats the two substantially bonded replicas B and C as economically rational actors and detects misconduct through blind replay before the primary opening, signatures over every consensus-critical field, an additional future audit after those signatures, objective timeouts, and self-contained fault evidence. In Layer 2, any observer may start a permissionless dispute during the PendingTicket challenge window and bisect the execution down to one Compute VM micro-step. Once objective evidence is accepted, branch-local consensus necessarily freezes collateral, suspends the credential, and executes the slash settlement without a vote or discretionary authority. A MatureTicket is generated only after the challenge window closes, every dispute terminates, and the state becomes stable.

Under the Testnet-23 Safety Profile, ALGO-3 is the sole source of ordering and liveness. A PALW ticket has no fork-choice weight, no transaction-ordering authority, and no transferable coinbase. A MatureTicket is finalized automatically as a zero-value conformance record. An ALGO-4 profile with non-zero ordering weight is outside the normative scope of this document and requires a separate safety argument, GHOSTDAG recalibration, independent audit, and hard fork. This fail-closed separation prevents uncalibrated useful-work economics from purchasing chain ordering.

1 Normative Scope and Deployment Profiles

This document normatively defines the PALW Core and the Testnet-23 Safety Profile. The output of the Core is a one-time MatureTicket proving that an execution commitment fixed before the lottery passed future randomness, NISA, bonded replay, and permissionless dispute.

1.1 Testnet-23 Safety Profile

Testnet-23 consensus parameters enforce the following values.

Item	Normative value
Ordering	ALGO-3 selected chain only
PALW fork-choice weight	0
PALW transaction-ordering authority	None
PALW coinbase / transferable reward	0
MatureTicket	Zero-value conformance record
Replica slash	Shadow slash ledger; real UTXO burn disabled
DNS / validator authority	Prohibited as consensus input
Compute Set	Genesis allowlist; nodes retain the complete artifact and Compute VM package

Consequently, a defect in a Testnet-23 PALW object cannot change the ALGO-3 fork choice, ordinary transaction state, or coinbase supply. BatchCommit, WinAnnounce, replica messages, openings, certificates, and disputes are carried as ALGO-3 control transactions or their consensus adjuncts. Maturity and finalization are deterministic ALGO-3 state transitions. The public testnet measures window reconstruction, deterministic execution, witnesses, replica assignment, disputes, reorganization rollback, IBD recovery, and resource budgets.

Version 0.6.1 is incompatible with the legacy PALW wire, the BLAKE2b/u128 target profile, and the OpeningAnnounce object. Testnet-23 uses a fresh genesis, a dedicated network identifier, a dedicated subnetwork byte, and a v0.6.1 RulesetID. Existing objects **MUST NOT** be reinterpreted in place.

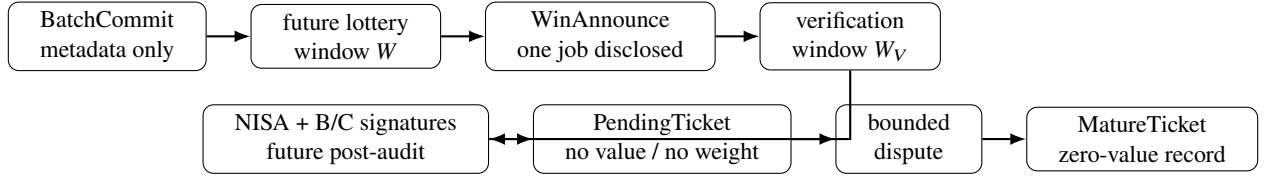


Figure 1: The only normative v0.6.1 flow. Future audit follows the B/C signatures; ordering, transaction effects, and transferable reward do not exist before the FinalCertificate and MatureTicket stages complete.

1.2 Value and Ordering Profiles

A Value Profile with transferable rewards remains disabled until the gates in Section 17 and a separate monetary-policy specification are satisfied. It treats B and C not as altruistic actors but as economically rational replicas that avoid losing substantial bonded collateral. Activation therefore requires a replica-rationality inequality using a conservative detection-probability floor, maximum bribe or ticket gain, irrecoverable slash, and credential-suspension cost. An Ordering Profile with non-zero fork-choice weight, ALGO-4 transaction execution, or block-template binding is not part of this document. An implementation that enables those functions is not v0.6.1-compliant and must be treated as a new protocol version subject to a new audit.

2 Design Principles

- P1. One execution, one commitment.** The authority sequence and leaf index enter the canonical input. Moving the same nominal job to another leaf, sequence, or BatchCommit produces a different execution.
- P2. Work precedes the lottery.** The projection commitment is accepted before the lottery window. Opening a commitment after learning the result without prior computation requires finding a hash preimage.
- P3. Lottery and result salts are separated.** The lottery salt revealed in WinAnnounce does not make the projection testable. The result salt remains hidden until after replica commitments.
- P4. Winner-only data availability.** Job bytes for losing leaves are not placed on chain. The selected job bytes are mandatory in the WinAnnounce carrier; an announcement without them is invalid.
- P5. A single pre-finality flow.** A Certificate creates a PendingTicket. A MatureTicket exists only after disputes terminate. Reward, ordering, and transaction effects do not exist earlier.
- P6. Branch-local state.** Sequence, first-wins state, capacity, bonds, disputes, tickets, nullifiers, and budgets are evaluated against the candidate block's parent state and roll back on reorganization.
- P7. Aggregate limits for every resource.** Consensus caps not only each object but also bytes, verifier cost, control transactions, active disputes, and roster utilization per selected-chain epoch.
- P8. Fail-closed activation.** If the Compute VM package, wire vectors, runtime conformance, or resource benchmark is absent, the profile's activation value remains zero.

3 Cryptographic Primitives and Canonical Encoding

The protocol hash H is domain-separated SHA3-512. Every preimage includes a protocol tag, wire version, network identifier, RulesetID, and fixed-order length-delimited fields. Target comparison interprets the first 256 output bits as a big-endian unsigned integer. Consensus objects **MUST NOT** contain floating-point values, host-endian integers, non-canonical map order, or unknown fields.

Authority and replica signatures use ML-DSA-87 with a distinct context string for each object class. The wire contract fixes leaf count, tree arity, odd-leaf handling, and path-direction bits for every canonical Merkle tree, so one root cannot ambiguously encode sequences of different lengths. Positive and negative vectors for hashing, signatures, Merkle proofs, fixed-point arithmetic, and rejection sampling are committed by the RulesetID. A ruleset that disagrees across two independent implementations cannot activate.

4 Homogeneous BatchCommit

4.1 Batch Descriptor and Canonical Input

A BatchCommit groups N executions of the same Compute Set s and the same shape σ . Heterogeneous batches are invalid in v0.6.1. The batch descriptor contains

$$D = (\text{network}, \text{ruleset}, h_A, \text{seq}, A_0, s, \sigma, N, r_{\text{reward}}). \quad (1)$$

Here h_A commits to the ML-DSA-87 authority public key, A_0 is a selected-chain ALGO-3 anchor, and r_{reward} is a reward-script commitment reserved for a future Value Profile. Testnet-23 fixes it to the zero script.

The authority sequence is not a worker-selected nonce. It must equal $\text{NextSequence}[h_A]$ in the parent state. Accepting a BatchCommit increments that sequence by exactly one, and a second transaction with the same pair is invalid. The anchor A_0 must be an ancestor within Δ_{anchor} ALGO-3 selected-chain blocks of the BatchCommit parent. Define

$$R_{\text{batch}} = H(\text{palw}/\text{batch-seed}/v1 \parallel D) \quad (2)$$

and, for leaf $j \in \{0, \dots, N-1\}$,

$$R_{\text{work},j} = H(\text{palw}/\text{work}/v3 \parallel R_{\text{batch}} \parallel \text{u32be}(j)). \quad (3)$$

The runtime inserts $R_{\text{work},j}$ into a protocol-defined input position. Duplicating the same prompt across leaves therefore does not duplicate an execution.

4.2 Execution Commitment and Salt Separation

Let the projection of execution j be

$$P_j = (r_{\text{exec},j}, r_{\text{out},j}, n_{\text{tok},j}, e_j, a_{\text{op},j}), \quad (4)$$

where $a_{\text{op},j}$ is the final value of an operation accumulator that commits to every semantic operation in order.

The worker generates mutually independent 256-bit values $\rho_{\text{res},j}$ and a batch-level lottery salt ρ_{lot} . Define

$$C_{\text{exec},j} = \text{H}(\text{palw/exec-commit/v1} \parallel j \parallel R_{\text{work},j} \parallel r_{\text{in},j} \parallel r_{\text{ctx},j} \parallel P_j \parallel \rho_{\text{res},j}), \quad (5)$$

$$C_{\text{lot}} = \text{H}(\text{palw/lot-commit/v1} \parallel \rho_{\text{lot}}). \quad (6)$$

The job leaf is

$$L_j = \text{H}(\text{palw/job-leaf/v1} \parallel j \parallel r_{\text{in},j} \parallel r_{\text{ctx},j} \parallel C_{\text{exec},j}), \quad (7)$$

and $r_{\text{jobs}} = \text{MerkleRoot}(L_0, \dots, L_{N-1})$. The batch identifier is

$$\text{BC} = \text{H}(\text{palw/batch-commit/v1} \parallel D \parallel r_{\text{jobs}} \parallel C_{\text{lot}}). \quad (8)$$

Revealing ρ_{lot} in WinAnnounce leaves $C_{\text{exec},j}$ hiding because it uses an independent $\rho_{\text{res},j}$. Conversely, C_{lot} independently binds the lottery salt and prevents post-window substitution.

4.3 Signature, Fee, and One-Shot Rules

The byte-level wire contract fixes the BatchCommit signing preimage. It covers at least

$$(\text{version, network, ruleset, } D, \text{BC, } r_{\text{jobs}}, C_{\text{lot}}, f_{\text{attempt}}, h_{\text{expiry}}). \quad (9)$$

The legacy form that omitted the commitment identifier itself from the signature is invalid.

A BatchCommit consumes only a non-refundable attempt fee:

$$f_{\text{attempt}} = f_0 + f_{\text{byte}}|\text{BatchCommit}| + f_q q_{\text{batch}}. \quad (10)$$

A fake leaf cannot dilute this fee. Work and replica-fee escrows are locked only when a WinAnnounce is accepted, so losing attempts do not immobilize large native-token balances.

Branch-local state contains $\text{batchState}[\text{BC}] \in \{\text{Committed, Announced, Expired}\}$ and $\text{lotCommitUsed}[C_{\text{lot}}]$. A lottery commitment can be registered only once on a branch. The lottery salt must remain fresh and secret from commitment through announcement. A BatchCommit can transition to WinAnnounce only once; any later announcement on the same branch is invalid. If no announcement is accepted within Δ_{announce} service opportunities after the lottery window ends, the batch becomes Expired and its active index and temporary metadata may be pruned.

For each shape, the trial count $q_{\text{job}}(s, \sigma)$ is derived by every node from the registered semantic program and opcode-cost vector. The batch count is

$$q_{\text{batch}} = N q_{\text{job}}(s, \sigma), \quad (11)$$

subject to $N \leq N_{\text{max}}$ and $q_{\text{batch}} \leq q_{\text{max}}^{\text{batch}}$. A worker-declared q is never a consensus input.

The global cap T_{max} satisfies

$$q_{\text{max}}^{\text{batch}} \frac{T_{\text{max}} + 1}{2^{256}} \leq \varepsilon_{\text{multi}}, \quad (12)$$

with initial $\varepsilon_{\text{multi}} = 2^{-10}$. Multiple wins inside one batch are therefore rare, bounding the non-linearity introduced by the one-ticket cap.

4.4 Uncomputed Leaves

An uncomputed leaf has no valid opening for $C_{\text{exec},j}$. If its trial wins, the worker cannot complete the post-announcement opening and loses the work escrow. Adding an uncomputed leaf does not add trials to any computed leaf, and one BatchCommit creates at most one ticket. The strategy cannot increase legitimate success probability or influence ordering; its cost is bounded by the attempt fee and winner escrow.

5 Canonical Execution, State Tree, and Compute VM

5.1 Artifacts Retained by Full Nodes

A v0.6.1 full node locally retains the model weights, tokenizer, semantic program, opcode table, shape table, state-tree schema, and conformance vectors for every active Compute Set. Interval witnesses do not embed weight tiles. Genesis and upgrade parameters cap the active-set count S_{max} and total weight bytes $B_{\text{weights}}^{\text{max}}$.

This removes the need to shrink checkpoint intervals to sub-matrix-multiplication granularity merely to satisfy a witness-tile limit. Weight availability is solved by package acquisition and hash verification before activation; block validity never depends on an online external fetch.

5.2 Semantic Operations and the Operation Accumulator

The Compute VM package completely specifies the typed inputs, integer ranges, overflow rules, read/write sets, canonical encoding, state transition, one-step verifier, and cost of every opcode. Its hash is ComputeVMID and is committed by the ruleset. Activation requires the package, positive and negative golden vectors, and two independent implementations.

Execution state contains at least the KV cache, activations, routing decisions, output, token counter, stop reason, and operation accumulator a_t . For every semantic operation o_t ,

$$a_{t+1} = H(\text{palw/op-acc/v1} \parallel a_t \parallel \text{enc}(o_t) \parallel \text{enc}(\text{reads}_t) \parallel \text{enc}(\text{result}_t) \parallel \text{enc}(\text{writes}_t)). \quad (13)$$

Scratch values overwritten before a checkpoint, discarded MoE paths, and padding operations all enter the accumulator. Omitting an operation that is not visible at the state boundary therefore still changes the checkpoint root.

5.3 Fixed-Operation Execution and Checkpoints

A shape executes exactly $\text{ops}(s, \sigma)$ cost units and satisfies

$$q_{\text{job}} = \frac{\text{ops}(s, \sigma)}{\kappa_q}, \quad q_{\text{cp}} = \frac{\text{ops}(s, \sigma)}{\kappa_{\text{cp}}}. \quad (14)$$

Both κ_q and κ_{cp} divide $\text{ops}(s, \sigma)$; partial quanta and partial intervals do not exist.

After natural decode termination, the runtime preserves the latched output and executes a deterministic padding program until the fixed operation budget is exhausted. Admission caps the padding-cost fraction of each shape at $u_{\text{pad}}^{\text{max}}$, and shortcut analysis covers the entire operation budget, including padding.

Let checkpoint roots be $D_0, \dots, D_{q_{\text{cp}}}$ and define

$$r_{\text{exec}} = \text{MerkleRoot}(D_0, \dots, D_{q_{\text{cp}}}). \quad (15)$$

An interval witness contains the pre- and post-checkpoints, touched state leaves, Merkle paths, and the opcode slice. Weights are read from the node's local active package.

6 Selected-Chain Randomness and Target

6.1 Batch-Specific Future Window

The sole randomness source is ALGO-3 blocks on the GHOSTDAG selected chain. Let B_c be the BatchCommit carrier hash. Define

$$\delta(\text{BC}) = \text{U256}(H(\text{palw/window-offset/v1} \parallel \text{BC} \parallel B_c)_{0..31}) \bmod w_{\text{spread}}. \quad (16)$$

Version 0.6.1 requires w_{spread} to be a power of two, eliminating modulo bias. The authority cannot know B_c before transaction inclusion and therefore cannot select an endpoint by grinding BatchID alone.

After $k + \delta(\text{BC})$ selected-chain ALGO-3 blocks, the next w blocks form an ordered MMR:

$$W = (B_c, A_{k+\delta}, \dots, A_{k+\delta+w-1}, r_{\text{MMR}}). \quad (17)$$

BatchCommits in the same carrier are spread over endpoints by BatchID, reducing the number of in-flight batches that one withheld boundary block can redraw.

6.2 WinAnnounce-Rate and Capacity Controller

The lottery target controls accepted WinAnnounces rather than post-challenge blocks. Let the controller anchor be (m_a, h_a, T_a) , let $m(h)$ be the accepted-WinAnnounce sequence through ALGO-3 selected-chain service height h , let I_W be the target WinAnnounce interval, and define

$$T_{\text{rate}}(h) = \text{clamp}\left(T_a 2^{\frac{h-h_a-(m(h)-m_a)I_W}{H_{\text{half}}}}, T_{\text{min}}, T_{\text{max}}\right) \quad (18)$$

using fixed-point integer arithmetic. Consensus requires $H_{\text{half}} \geq 10\Delta_{\text{announce}}$, making the controller slow relative to the target-freeze-to-announcement dead time.

Let U_{ref} be the reserved-slot ratio at the lottery-window endpoint. The capacity clamp is

$$T_{\text{cap}}(U) = \begin{cases} T_{\text{max}}, & U \leq U_{\text{soft}}, \\ T_{\text{max}} \frac{U_{\text{max}} - U}{U_{\text{max}} - U_{\text{soft}}}, & U_{\text{soft}} < U < U_{\text{max}}, \\ 0, & U \geq U_{\text{max}} \end{cases} \quad (19)$$

The ticket target is

$$T_{\text{ticket}} = \min(T_{\text{rate}}(h), T_{\text{cap}}(U_{\text{ref}})). \quad (20)$$

If $U_{\text{ref}} \geq U_{\text{max}}$, an admission-closed flag is also set and a WinAnnounce is rejected even if its hash is zero. Saturated replica capacity can never cause the target to loosen.

The lottery endpoint freezes

$$\text{ref}_T = (\text{blockID}, \text{stateRoot}, \text{controllerVersion}, \text{fixedPointID}, U_{\text{ref}}). \quad (21)$$

Every WinAnnounce, Certificate, PendingTicket, and MatureTicket signature preimage binds ref_T and T_{ticket} . No later step consults a live target.

6.3 Private Trials and the One-Time WinAnnounce

Define

$$R_{\text{job}} = H(\text{palw/job-randomness/v5} \parallel \text{BC} \parallel W), \quad (22)$$

$$\nu_{\text{lot}} = H(\text{palw/trial-key/v3} \parallel \rho_{\text{lot}}), \quad (23)$$

$$X_i = H(\text{palw/trial/v3} \parallel R_{\text{job}} \parallel \text{BC} \parallel \text{u64be}(i) \parallel \nu_{\text{lot}}). \quad (24)$$

The first 256 SHA3-512 output bits are interpreted as a big-endian U256. The worker may announce any index $0 \leq i < q_{\text{batch}}$ with $X_i \leq T_{\text{ticket}}$. A batch yields at most one ticket, and verification randomness does not depend on the winning index or job leaf, so selecting among several wins cannot redraw replicas or audit samples.

The index maps uniquely to a job and quantum:

$$j = \left\lfloor \frac{i}{q_{\text{job}}} \right\rfloor, \quad u = i \bmod q_{\text{job}}. \quad (25)$$

A WinAnnounce contains

$$\text{WA} = (\text{BC}, h_A, \rho_{\text{lot}}, i, j, u, \text{ref}_T, T_{\text{ticket}}, r_{\text{jobBytes}}, \pi_{\text{leaf}}, E_w, E_{\text{rep}}). \quad (26)$$

The authority signature covers the entire canonical WinAnnounce, network, RulesetID, and escrow outputs. Its carrier must include the selected public input and context bytes in a consensus adjunct. Full nodes immediately verify the C_{lot} opening, job-leaf path, shape-derived q , index mapping, winning inequality, signature, escrows, epoch win cap, and capacity permit.

WinAnnounces in one block are processed in canonical transaction-index order. Every accepted transition atomically reserves capacity and budget. A transaction lacking a capacity permit cannot be included as a valid transition, so it neither locks escrow nor acquires first-wins state. The branch-local transition

$$\text{announced}[\text{BC}] : \perp \longrightarrow H(\text{WA}) \quad (27)$$

accepts only the first valid transaction.

6.4 Verification Window and Roster Reference

The replica roster root r_{slots} is frozen from branch state at the lottery-window endpoint, not at the WinAnnounce carrier. The worker therefore cannot time the announcement to choose roster composition.

After k_2 blocks following the WinAnnounce carrier, the next w_2 ALGO-3 blocks form W_V . Define

$$R_{\text{rep}} = H(\text{palw/replica-randomness/v5} \parallel \text{BC} \parallel \text{carrier}(\text{WA}) \parallel W_V \parallel r_{\text{slots}}), \quad (28)$$

$$R_{\text{audit}} = H(\text{palw/audit/v5} \parallel \text{BC} \parallel \text{carrier}(\text{WA}) \parallel W_V). \quad (29)$$

The winning index, job-bytes hash, projection, and result salt are not inputs to these values.

7 Data Availability and Consensus Resource Budgets

Losing job bytes are never placed on chain. The WinAnnounce carrier contains the selected job bytes. If the primary opening, two replica reveals, post-audit reveals, and FinalCertificate span several control transactions, the total canonical evidence for one ticket must remain at or below $B_{\text{pending}}^{\text{max}}$. The FinalCertificate transition binds their hashes and completion of availability, and the PendingTicket references that FinalCertificateID. Block validity cannot depend on an external online fetch.

Each ALGO-3 block has a PALW control-payload limit B_{ctrl} separate from ordinary transaction capacity. A deterministic weighted round-robin calendar assigns service slots to object classes with deadlines. A scheduled class may use up to B_{ctrl} in that block; unused space may be borrowed, but the class service clock advances only under the canonical calendar. Ordinary fee-market congestion and traffic in another class cannot consume reserved service opportunities. BatchCommits have no response deadline and use residual space.

The largest WinAnnounce, replica commitment, primary opening, replica reveal, post-attestation audit reveal, ReplicaEvidence reveal, challenge reveal, dispute response, and ChallengeOverflowSignal must each fit within B_{ctrl} . An evidence format that does not fit is not activatable and must not be silently split across blocks under an undefined timeout rule. The Testnet-23 cycle is sixteen ALGO-3 blocks. Every listed deadline class receives at least one slot per cycle, and dispute response and ReplicaEvidence receive additional reserved slots. The exact map is committed by the parameter package and golden vectors.

A budget epoch has non-borrowable class reserves $B_{\text{epoch},c}$ and $V_{\text{epoch},c}$ whose sums fit under the global caps. Other classes and BatchCommits cannot consume a deadline class's epoch reserve. Each class maintains a deterministic fair queue ordered by (deadline, scopeID, objectKind, objectHash), where scopeID is the relevant BatchID, PendingTicketID, or gameID. A calendar slot services the queue head and advances only that object's deadline clock. One ticket cannot consume a slot while another honest ticket's clock advances.

A class-specific service turn exists only when the calendar schedules the class and its reserved byte/cost budget can admit that object's maximum envelope. Thus the protocol's own aggregate cap cannot advance an honest response clock while making inclusion impossible.

There are exactly two time axes. Lottery windows, announcement expiry, challenge windows, retention, and stability depth are counted in ALGO-3 selected-chain height. Slashable response deadlines are counted in service turns allocated to the specific object. The Ruleset fixes the cycle length g_{cycle} and each class's maximum slot gap g_c , so a deadline of n opportunities has a reproducible upper bound of ng_c selected-chain blocks. Producer-local clocks, DAA score, and external epochs are not used.

For every selected-chain budget epoch E_B , consensus enforces

$$\sum \text{adjunctBytes} \leq B_{\text{epoch}}, \quad (30)$$

$$\sum \text{VerifierCost} \leq V_{\text{epoch}}, \quad (31)$$

$$\# \text{BatchCommit} \leq C_{\text{batch}}, \quad (32)$$

$$\# \text{WinAnnounce} \leq C_{\text{win}}, \quad (33)$$

$$\# \text{newDispute} \leq C_{\text{dispute}}. \quad (34)$$

An object is invalid if adding it would exceed either a per-object cap or an epoch cap. The product of permissive independent maxima therefore cannot create unbounded bandwidth, CPU, or retention.

An object's timeout clock advances only on its own fair-queue service turns. Heights where it is not assigned a turn do not count. Continued exclusion during assigned turns is classified as targeted hash-lane censorship, not ordinary transaction congestion or PALW-class starvation.

8 Layer 0: NISA

8.1 Unique Samples

NISA indices are drawn without replacement from R_{audit} by rejection sampling. Replica B and C indices are then drawn in fixed order while skipping all NISA and earlier replica indices. Therefore

$$\tau_{\text{eff}} = \tau + 2\kappa_r + 2\kappa_p \quad (35)$$

random intervals are pairwise distinct by construction. Here κ_p is the number of additional post-attestation intervals required from each replica after it has signed. The first and last boundary intervals are additional checks.

8.2 Opening and Witness

The primary opening contains the selected leaf's $(r_{\text{in}}, r_{\text{ctx}}, P, \rho_{\text{res}})$, Merkle path, checkpoint roots, and NISA witnesses. A full node verifies

- (i) the leaf path and C_{exec} opening;
- (ii) $R_{\text{work},j}$ and the initial state;
- (iii) T_{ticket} and the WinAnnounce winning trial;
- (iv) κ_{cp} operations in every sampled interval; and
- (v) the operation accumulator, first and last boundaries, and output projection.

A deleted candidate target or a live target at finalization time must never be consulted.

An interval witness is

$$\Pi_t = (D_t, D_{t+1}, L_t, \pi_t^{\text{state}}, \pi_t^{\text{exec}}, \text{opcodeSlice}_t), \quad (36)$$

with weights read from the active Compute Set package. Witness size and one-interval verifier cost are frozen at shape admission.

9 Layer 1: Bonded Replica Verification

9.1 Roster, Capacity, and Canonical Assignment

A replica lease records a credential, Compute Set, shape, bond, performance slots, correctness-lien slots, activation height, and exit state. Only active leases aged at least Δ_{age} at the lottery-window endpoint enter the roster root.

When a WinAnnounce is accepted, the network provisionally reserves two global capacity tokens. The global and credential reserved ratios satisfy

$$U_{\text{global}} = \frac{\text{reservedSlots}}{\text{totalActiveSlots}} \leq U_{\text{max}}, \quad U_c = \frac{\text{reservedSlots}_c}{\text{activeSlots}_c} \leq U_{\text{max}}. \quad (37)$$

The global ratio is checked at WinAnnounce acceptance and the credential ratio when B/C are assigned. A transition exceeding either bound is rejected. Initial $U_{\text{max}} = 1/4$. A credential's nominal performance-slot capacity $S_c = \lfloor B_{\text{total}}/B_{\text{perf}} \rfloor$ is rederived by nodes rather than self-declared, and must satisfy the collateral inequality. Define $U_c = \text{reservedPerfSlots}_c/S_c$. Because every honest credential retains at least $1 - U_{\text{max}}$ free fraction, an attacker who deliberately keeps all slots idle receives only a bounded increase in free-slot share.

At the end of W_V , nodes derive a roster permutation from R_{rep} , exclude the primary credential, and select the first branch-local free performance slots for B and C. If multiple assignments settle in one block, canonical order is the lexicographic order of $(W_V \text{ end height, BC})$. The provisional tokens atomically convert into B/C reservations. If two distinct credentials cannot be selected, the tokens are released and the ticket path terminates fail-closed; the protocol never degrades to a shorter list.

9.2 Two-Stage Bond Reservation

Each selected replica reserves B_{perf} until FinalCertificate creation or an objective failure. When B or C signs CB, the embedded B_{corr} becomes a non-releasable correctness sub-lien. After FinalCertificate creation, $B_{\text{perf}} - B_{\text{corr}}$ is released and B_{corr} remains as a correctness lien until maturity or rejection.

For every authority,

$$\sum_{\text{active assignments}} B_{\text{perf}} + \sum_{\text{pending certificates}} B_{\text{corr}} \leq B_{\text{rep}}. \quad (38)$$

A proven false Certificate slashes its correctness lien, suspends the credential, and freezes other pending Certificates backed by the same bond. In a Value Profile, hard-fork parameters require B_{corr} to be at least the maximum ticket value and require $p_{\text{det}}^{\min} S_R^{\min} > G_R^{\max}$. Testnet-23 does not burn transferable UTXOs, but applies shadow amounts, credential suspension, capacity removal, and freeze transitions in branch-local state.

An exit request immediately removes a lease from new roster eligibility, but bond release is forbidden until all performance reservations, correctness liens, and pending SlashReports are zero and $R_{\text{arch}} + \Delta_{\text{stable}}$ has elapsed since the final signed Certificate. Collateral cannot escape during the evidence horizon.

No-show evidence is self-contained: it identifies the assignment, required object class, assigned service turns, deadline, and unconsumed response ID. Reaching $N_{\text{ns}}^{\max}$ no-shows in the rolling epoch suspends the lease until cooldown and capacity top-up complete.

9.3 Economic Premise and Replica Rationality

A Value Profile assumes that B and C are economically rational actors that avoid losing substantial bonded collateral. This is not a declaration that replicas never cheat. It is an activation condition under which multiple detection routes and mandatory enforcement satisfy

$$G_R^{\max} < p_{\text{det}}^{\min} S_R^{\min}. \quad (39)$$

Here $G_R^{\max}$ is the maximum bribe and ticket-related gain available to a replica, $p_{\text{det}}^{\min}$ is a conservative detection floor measured from mandatory audits and watcher exercises, and $S_R^{\min}$ is the minimum irrecoverable loss for that fault after subtracting any bounty recoverable by the coalition from slash, credential suspension, lost capacity, and exit delay. A Value Profile cannot activate unless measured and worst-case values satisfy this inequality. Testnet-23 performs no real burn, so it does not prove the rationality assumption; it tests completeness of detection and shadow enforcement.

9.4 Evidence Completeness and Signed Fields

Every replica obligation registers an AssignmentID, role, object kind, expected body hash, and service-turn deadline in branch-local state. By the deadline either a response with a canonical outer signature exists or an objective timeout exists. There is no ambiguous silence state.

A replica envelope is

$$\text{REnv}_R = (\text{network}, \text{ruleset}, \text{branchAnchor}, \text{AssignmentID}, \text{role}, \text{objectKind}, \text{bodyHash}, \text{deadline}, \sigma_R). \quad (40)$$

An invalid outer signature can be fabricated by anyone and therefore must never cause a slash by itself; it is rejected as a response and may eventually produce no-show timeout evidence. Conversely, if the outer signature is valid and the signed body is non-canonical, contradictory, or inconsistent with branch facts, it is objective protocol-fraud evidence attributable to the signer.

The CertificateBody signed by B and C binds at least

$$\begin{aligned} \text{CB} = & (\text{network}, \text{ruleset}, \text{AssignmentID}, \text{BC}, \text{H(WA)}, \\ & \text{ref}_T, T_{\text{ticket}}, \text{rosterRoot}, \text{jobLeafHash}, C_{\text{exec}}, \\ & \text{H}(P), r_{\text{exec}}, r_{\text{NISA}}, r_B, r_C, h_A). \end{aligned} \quad (41)$$

Only fields independently rederived by full nodes appear in this preimage. B and C sign the same $\text{H}(\text{CB})$ under role-separated contexts, preventing substitution of target, roster, assignment, projection, execution root, or evidence roots. The system-derived PendingTicket later refers to FinalCertificateID, so no circular ticket identifier appears in this preimage.

9.5 Execution-Time-Dependent Deadline

The shape table freezes a conservative ALGO-3-service-block execution bound $T_{\text{exec}}(s, \sigma)$ from conformance benchmarks. The replica-commitment deadline is

$$\Delta_{\text{rs}}(s, \sigma) = \lceil \alpha_{\text{exec}} T_{\text{exec}}(s, \sigma) \rceil + \Delta_{\text{net}}, \quad \alpha_{\text{exec}} \geq 3. \quad (42)$$

The protocol does not use a fixed deadline of a few seconds. Primary-opening and replica-reveal deadlines are likewise derived from shape bounds and control-band propagation budgets.

9.6 Blind Commitment, Opening, and Certificate

B and C independently start execution from the published job bytes and, before the primary opening, submit

$$\text{com}_R = \text{H}(\text{palw/replica-commit/v4} \parallel \text{BC} \parallel h_R \parallel P_R \parallel \rho_R). \quad (43)$$

After both commitments are accepted, the primary opens P_A, ρ_{res} and NISA evidence. B and C then reveal (P_R, ρ_R) and replica-specific interval witnesses.

A projection mismatch does not create a Certificate. It enters pre-PendingTicket adjudication. Objective bisection identifies the losing party, and if one replica is malicious the next canonical slot replaces it so that a single replica cannot destroy an honest winning path. A Certificate exists only if

$$P_A = P_B = P_C \quad (44)$$

and binds all signatures, commitment openings, NISA, replica witnesses, target reference, roster root, and bond reservations.

A no-show slash is valid only if the deadline contained the required number of PALW service opportunities and the class control band was not paused by capacity exhaustion. Otherwise the assignment is deferred without slashing. Sustained targeted exclusion is measured separately as hash-lane liveness failure.

9.7 Future Audit after Replica Signatures

Once $P_A = P_B = P_C$ and the initial witnesses pass, B and C sign CB. Let A_{SC} be the first block on a branch in which both signatures coexist and define

$$\text{SCID} = \text{H}(\text{palw/signed-certificate/v1} \parallel \text{CB} \parallel \sigma_B \parallel \sigma_C). \quad (45)$$

A certificate-specific offset and a future ALGO-3 window W_P after A_{SC} derive

$$R_{\text{post}} = \text{H}(\text{palw/post-replica-audit/v1} \parallel \text{SCID} \parallel W_P), \quad (46)$$

$$S_{\text{post},R} = \text{Sample}(\text{H}(R_{\text{post}} \parallel h_R), \kappa_P). \quad (47)$$

The B and C post-audit sets skip every NISA, initial-replica, and earlier post-audit index. Each replica must reveal these interval witnesses before its deadline. Thus a replica that irrevocably signs a trace or ticket tailored only to the initial samples receives an additional objective detection route and an additional no-show route after the signature is fixed.

This future audit is not a cryptographic proof of independent execution. A replica that receives a correct trace from the primary is indistinguishable from one that replayed locally. It does, however, detect false execution, inconsistent execution roots, unavailable trace state, and refusal after signature. If either replica refuses or fails the post-audit, consensus freezes that replica's performance/correctness sub-lien and advances to the next canonical slot. A replacement creates a new CB and signatures and derives a new SCID and future W_P ; it cannot reuse the old SCID or known samples. Escalation is bounded by E , after which the path terminates fail-closed without primary fault.

The first block in which both post-audits pass creates the system-derived

$$\text{FCID} = \text{H}(\text{palw/final-certificate/v1} \parallel \text{SCID} \parallel r_{\text{postEvidence}}), \quad (48)$$

and only this FinalCertificate may create a PendingTicket. The W_P endpoint is spread by an offset derived from SCID and the carrier hash; a Value Profile requires withholding calibration.

9.8 Objective Evidence and Deterministic Slashing

Canonical evidence is uniquely identified by

$$\text{OffenseID} = \text{H}(\text{palw/replica-offense/v1} \parallel \text{kind} \parallel \text{branchAnchor} \parallel \text{AssignmentID} \parallel h_R \parallel \text{evidenceDigest}), \quad (49)$$

and advances the one-shot state

$$\text{FaultState}[\text{OffenseID}] \in \{\text{absent}, \text{frozen}, \text{settled}\}. \quad (50)$$

In the same state transition that accepts valid evidence, consensus freezes the relevant B_{perf} or B_{corr} , suspends the credential, sets its selection capacity to zero, and freezes other pending Certificates backed by the same bond. After Δ_{stable} , it automatically executes slash, reporter bounty, and burn, or the Testnet-23 shadow settlement. No validator vote, committee approval, or operator discretion exists. Reorganization rolls evidence and settlement back through the same undo record. Each (AssignmentID, role, phase) has one terminal fault state; multiple evidence objects proving the same fault are deduplicated as aliases. Performance and correctness phases use separate liens, but aggregate slash in each phase cannot exceed its reservation and one fault cannot pay multiple bounties. If several kinds are proven before settlement, a severity priority fixed by the parameter package merges them uniquely.

Timeout evidence is generated by a system transition and needs no reporter. Anyone may submit two signed conflicting objects, a commitment mismatch, an invalid signed witness, or a CertificateBody inconsistent with branch facts through the replica-evidence transaction class. Commit-reveal binds bounty ownership to the first valid commitment over evidence hash and reporter key, preventing mempool copying. The evidence class has reserved calendar and epoch capacity and does not expire before collateral can be released.

Fault	Objective detection route	Enforcement	Ticket effect
No-show / reveal refusal	Assigned service turns and unconsumed response ID	S_{ns} , suspension threshold	Replacement or fail closed

Fault	Objective detection route	Enforcement	Ticket effect
Authenticated malformed body	Valid outer signature plus canonical-decode failure	S_{proto} , immediate suspension	Invalid object; no PendingTicket
Equivocation	Two distinct valid signatures for one assignment/role/kind	S_{proto} , capacity freeze	Invalid Certificate
Commitment-opening mismatch	Signed reveal inconsistent with prior commitment	S_{proto}	Replacement and readjudication
Invalid initial/post-audit witness	Merkle or Compute VM failure in a signed witness	S_{perf} , suspension	No FinalCertificate
Projection-mismatch loss	Final pre-PendingTicket bisection result	S_{mis}	Preserve honest party and replace loser
Invalid ticket binding	Signed CB target/roster/assignment fields conflict with branch facts	S_{proto} or B_{corr}	No system ticket
False-execution Certificate	Final permissionless-dispute or future-audit result	Both B/C B_{corr} , suspension	Reject PendingTicket

Unauthenticated invalid bytes, an invalid signature fabricated by a third party, or an object valid only on another branch are not slash evidence. Evidence is accepted only when the reporting branch contains the assignment and signing obligation and verifies either the offender's valid signature or objective timeout.

10 Layer 2: Permissionless Dispute

10.1 PendingTicket

The FinalCertificate system transition creates

$$\text{PT} = (\text{BC}, \text{H}(\text{WA}), \text{ref}_T, T_{\text{ticket}}, \text{H}(P), r_{\text{exec}}, r_{\text{certificate}}, r_{\text{evidence}}, h_A). \quad (51)$$

A PendingTicket has no reward, fork-choice weight, or transaction-execution right. The challenge window remains open for $\Delta_{\text{challenge}}$ ALGO-3 selected-chain blocks after the PendingTicket creation block. The legacy lottery-end expiry is not reused as dispute lifetime.

Certificate-pipeline time is budgeted independently:

$$\Delta_{\text{cert}} = k_2 + w_2 + E\Delta_{\text{rs}} + \Delta_{\text{open}} + \Delta_{\text{reveal}} + k_3 + w_3 + \Delta_{\text{post}} + \Delta_{\text{adj}}^{\text{max}}, \quad (52)$$

where E is the escalation-slot cap selected only after all other phases have been subtracted.

10.2 Bounded Challenge Set

A challenge begins with commit-reveal. The commitment binds the ticket, interval, claim hash, reporter key, and nonce. It expires and loses bounty priority unless revealed within Δ_{cr} . Identical (PT, interval, postClaim) values are deduplicated by claim ID.

For each ticket,

$$D_{\text{active}} \leq D_{\text{max}}^{\text{active}}, \quad D_{\text{total}} \leq D_{\text{max}}^{\text{total}}, \quad B_{\text{queued}} \leq B_{\text{max}}^{\text{queue}}, \quad (53)$$

and the epoch dispute-verifier budget is also capped. Challenge commitments close Δ_{cr} before the challenge-window endpoint so that no unrevealed commitment remains at closure.

When $D_{\text{total}} = D_{\text{max}}^{\text{total}}$, one fixed-size ChallengeOverflowSignal is admissible through a dedicated service slot outside the normal challenge cap and C_{dispute} . The signal contains only ticket ID, reporter key, fresh nonce, and B_{overflow} bond. A timely signal causes a fail-closed ChallengeOverflowReject transition. The protocol burns $B_{\text{overflow}} + E_w$, returns existing challenge bonds, and applies no correctness slash to replicas. A future Value Profile must require this combined amount to exceed maximum ticket value plus permitted grief value and must separately audit the liveness trade-off. Testnet-23 uses the same safety-first rejection rule. Challenge-slot occupation therefore cannot produce a false maturity.

10.3 Bisection and One-Step Adjudication

A game explicitly records the agreed pre-state, primary post-state, challenger post-state, turn, and deadline. The midpoint is $\lfloor (\text{lo} + \text{hi})/2 \rfloor$, and the protocol uniquely selects the half on which the claims differ. After reaching one checkpoint interval, the same rule is applied over Compute VM opcode indices until a full node executes one micro-step.

Invalid moves, wrong turns, malformed witnesses, and missed deadlines have normative losing transitions. One proven defect rejects the PendingTicket and deterministically settles work escrow and false-Certificate liens. A false challenge pays defender response cost from the challenger bond.

10.4 State-Dependent Maturity

A PendingTicket transitions to MatureTicket only when

$$\text{challengeClosed} \wedge D_{\text{active}} = 0 \wedge D_{\text{queued}} = 0 \wedge \text{settledStable}. \quad (54)$$

Maturity is not determined by one fixed height. Define

$$\text{MT} = (\text{H}(\text{PT}), r_{\text{maturityState}}, r_{\text{disputeSet}}, \text{nullifier}, r_{\text{reward}}) \quad (55)$$

and

$$NF_{\text{ticket}} = H(\text{palw/ticket-nullifier/v1} \parallel H(PT)). \quad (56)$$

The nullifier is consumed once in branch-local state.

In Testnet-23, the first ALGO-3 block whose state satisfies the maturity condition automatically appends the zero-value record to the accumulator and consumes the nullifier. There is no FinalizeTicket transaction, online authority signature, or short use window. Targeted denial of service against the winner cannot destroy a mature ticket.

11 Fees, Escrows, and Settlement

The attempt fee is consumed when the BatchCommit is accepted and never refunded. A WinAnnounce creates work escrow E_w and replica-fee escrow E_{rep} . The work escrow makes “commit without computation and buy replica capacity only after winning” an expected-loss strategy.

A Value Profile must satisfy

$$E_w \geq \gamma_{\text{grief}}(C_B + C_C + C_{\text{include}} + C_{\text{verify}} + V_{\text{ticket}}), \quad \gamma_{\text{grief}} > 1. \quad (57)$$

Testnet-23 has no transferable value but simulates the same equation with test credits and capacity accounting; epoch caps bound damage from accepted fake wins.

Terminal state	Attempt fee	Replica escrow	Work escrow / ticket
No WinAnnounce	Consumed	None	None
Invalid WinAnnounce	Consumed	Transaction rejected	Transaction rejected
Replica no-show; escalation succeeds	Consumed	Pay completed replicas; apply shadow slash first	Continues
Primary opening absent or invalid	Consumed	Pay committed replicas; refund remainder	Full burn or shadow burn
Primary loses mismatch adjudication	Consumed	Pay honest replicas	Full burn; no PendingTicket
Replica loses mismatch adjudication	Consumed	Pay honest replicas, including replacement	Continues; replica lien slashed
FinalCertificate fails without primary fault	Consumed	Pay completed work; refund remainder	Refunded
Post-audit or challenge proves execution false	Consumed	Already paid	Challenger bounty; burn remainder; reject ticket
False challenge or timeout	Consumed	Continues	Pay defender cost from challenger bond; burn remainder
Challenge-overflow rejection	Consumed	Pay completed work	Reject; burn $B_{\text{overflow}} + E_w$; return existing challenge bonds
All challenges resolved; MatureTicket	Consumed	Pay B/C	Predetermined reward in Value Profile; zero in Testnet-23
Automatic MatureTicket finalization	Consumed	Complete	System transition consumes nullifier

Every terminal transition has a branch-local undo record. Slash, refund, and maturity before stability depth roll back under reorganization. Subsidy, fee share, and bond multipliers in a Value Profile are consensus constants; changing them requires a hard fork and independent economic audit.

12 Storage, Pruning, and Full-Node Cost

Winner-only data and aggregate budgets prevent attempt rate from increasing retained job bytes linearly. A BatchCommit carries near-fixed metadata, a WinAnnounce carries one job, and a PendingTicket carries one opening, one FinalCertificate, and one post-audit evidence set.

Public job bytes, openings, witnesses, and dispute data are retained for at least

$$R_{\text{arch}} \geq \Delta_{\text{challenge}} + \Delta_{\text{game}}^{\text{max}} + \Delta_{\text{stable}} \quad (58)$$

ALGO-3 selected-chain blocks. Here $\Delta_{\text{game}}^{\text{max}}$ is the block bound derived from service-turn limits and class maximum gap g_c . Non-archival nodes may prune them afterward but retain BatchCommit, WinAnnounce, PendingTicket, MatureTicket, dispute result, and accumulator roots. Full verification of older useful work requires archival data, and snapshot bootstrap explicitly states this assumption.

The normal full-node path retains weights and verifies the selected-chain windows, one winning hash, $\tau + 2 + 2\kappa_r + 2\kappa_p$ intervals, ML-DSA signatures, bond state, dispute state, and accumulator transition. It never performs a full-model replay.

13 Single Global Lane and Compute Set Admission

All admitted Compute Sets share one WinAnnounce-rate target, one fee rule, and one evidence format. Model-specific targets, quotas, premiums, and validator allocations do not exist. Work quantity contributes linearly to win probability through q ; post-discovery MatureTicket value does not depend on q .

An activation package includes at least

- model, tokenizer, template, and weights hashes;
- exact integer or fixed-point arithmetic;
- ComputeVMID, opcode-cost vector, state schema, and operation-accumulator rule;
- homogeneous shape table, q_{job} , κ_{cp} , T_{exec} , and padding cap;
- positive and negative conformance vectors;
- shortcut analysis covering dead paths and padding;
- results from two independent implementations; and
- a secret-handling profile: production builds prohibit Copy/Clone for ρ_{lot} , ρ_{res} , and trial keys and zeroize them after use.

A Compute Set manifest and artifact hashes are fixed at least Δ_{setLead} service blocks before explicit activation. Every reference node must possess the complete package and pass its vectors. A node missing the local artifact must not treat an object using that Set as verified. An external online service is never a source of truth.

14 Security Analysis

14.1 Partial Computation

Let $x \in [0, 1]$ be the fraction of semantic-operation cost an attacker omits. Unless a semantically equivalent shortcut exists, the operation accumulator makes the omission create at least one defective checkpoint interval. If $d(x)$ is the defective-interval fraction, the probability that all distinct $\tau_{\text{eff}} = \tau + 2\kappa_r + 2\kappa_p$ samples miss defects is bounded by

$$\Pr[\text{miss}] \leq (1 - d(x))^{\tau_{\text{eff}}}. \quad (59)$$

A genuine shortcut is a work-pricing failure rather than execution falsification and is handled by Compute Set admission and the padding cap.

Let c be honest execution cost, V the value of a winning path, f the attempt fee, and S the loss on detection. A coarse upper bound on partial-computation expected value is

$$\text{EV}(x) \leq pV(1 - d(x))^{\tau_{\text{eff}}} - (1 - x)c - f - S(1 - (1 - d(x))^{\tau_{\text{eff}}}). \quad (60)$$

A Value Profile must use measured $d(x)$ and collateral to demonstrate that every $x > 0$ underperforms honest execution. Testnet-23 fixes value to zero and prevents this economic assumption from affecting chain ordering.

14.2 Fake Batch Leaves and One-Shot Probability

An uncomputed leaf cannot be opened if it wins. The trial set of every computed leaf is unchanged by the presence of fake leaves, and one BatchCommit creates one ticket. A fake fraction ϕ therefore lowers the fraction of usable wins to $(1 - \phi)$ and never increases legitimate work trials.

Even if several winning indices exist, R_{rep} and R_{audit} are independent of index and leaf. The worker may choose only which already-earned job receives the one ticket; it cannot redraw samples or roster.

14.3 Replica Capture and Utilization Bound

Let β be the attacker's total bonded-slot share. Assume honest slot utilization is at most U_{max} and the attacker keeps all slots free. The free-slot share is bounded by

$$\beta_{\text{free}} \leq \frac{\beta}{\beta + (1 - \beta)(1 - U_{\text{max}})}. \quad (61)$$

This bound follows because consensus enforces a utilization cap on every credential. Capturing both B and C is approximately the without-replacement analogue of β_{free}^2 . A false ticket must additionally pass NISA and replica witnesses and survive the challenge window. Security reports therefore use measured free-slot concentration and bribable share rather than average bond share alone.

14.4 Detection Floor and Slash Adequacy for Replica Misconduct

If an execution defect occupies random-interval fraction d , the mandatory NISA, initial replica witnesses, and future audit after signatures give the detection floor

$$p_{\text{det}}(d) \geq 1 - (1 - d)^{\tau + 2\kappa_r + 2\kappa_p}. \quad (62)$$

Permissionless disputes, objective timeouts, and signed protocol-fraud evidence add further routes. Structural ticket fraud, equivocation, and a validly signed malformed body are detected with probability one by full-node rederivation. The protocol does not claim probability-one detection for unsampled portions of false execution.

A Value Profile must measure an adversarial lower bound on d , watcher coverage, and maximum bribery and satisfy the replica-rationality inequality for every supported fault class. Slash adequacy uses irrecoverable loss after reporter bounty and coalition recovery, not gross bond. No discretionary path may accept evidence yet omit the slash transition.

14.5 Withholding

Batch endpoints and post-attestation-window endpoints are spread using carrier hash with BatchID or SCID, respectively. ALGO-3 withholding bias is not eliminated, but aggregate budgets and spreading bound the number of batches one boundary choice can redraw. Testnet-23 measures

$$\max \mathbb{E}[\text{bias benefit}] < \mathbb{E}[\text{withholding cost}]. \quad (63)$$

Because PALW has zero ordering weight in this profile, the benefit is limited to zero-value telemetry.

14.6 Control-Loop Stability

The controller observes accepted WinAnnounces and removes Certificate and challenge delay from the feedback loop. It simultaneously enforces $H_{\text{half}} \geq 10\Delta_{\text{announce}}$, the capacity clamp, epoch win cap, and target clamp. Activation requires simulation under randomized supply shocks, replica outages, and capacity saturation without sustained oscillation.

15 Initial Testnet-23 Parameters

These are starting values for the zero-value Safety Profile, not a Value Profile. Horizons labelled “ALGO-3 blocks” are selected-chain heights; response deadlines labelled “service opportunities” are class-calendar slots.

Parameter	Initial value	Purpose
Hash / target	SHA3-512 / U256	First 256 output bits, big-endian
k, w	64, 256	Future lottery window
Δ_{anchor}	4096 ALGO-3 blocks	Execution-freshness bound
w_{spread}	256	Spread shared-endpoint bias
k_2, w_2	64, 256	Post-announcement verification window
k_3, w_3	64, 64	Future audit window after B/C signatures
N_{max}	1024	Homogeneous-batch leaf cap
$q_{\text{max}}^{\text{batch}}$	2^{24}	Fee, loop, and index cap
ϵ_{multi}	2^{-10}	One-ticket non-linearity bound
Δ_{announce}	1024 ALGO-3 blocks	Announcement deadline after lottery endpoint (about 128 s)
C_{batch}	4096 / epoch	BatchCommit state-growth cap
C_{win}	4 / epoch	Replica-pipeline admission cap
C_{dispute}	16 / epoch	New-dispute admission cap
$B_{\text{job}}^{\text{max}}$	512 KiB	Selected job bytes in WinAnnounce
$B_{\text{pending}}^{\text{max}}$	8 MiB	Opening, initial/post-audit witnesses, FinalCertificate
B_{ctrl}	8 MiB / ALGO-3 block	Maximum scheduled PALW-class payload
g_{cycle}	16 ALGO-3 blocks	Class-specific service calendar
E_B / B_{epoch}	1024 blocks / 64 MiB	Aggregate adjunct cap (average 64 KiB/block)
$B_{\text{epoch},c}$	3/36/16/9 MiB	Job / ticket evidence / dispute / metadata reserves
τ, κ_r, κ_p	16, 4, 4	Distinct NISA, initial-replica, and post-signature samples
U_{max}	1/4	Roster-utilization hard cap
U_{soft}	1/8	Utilization where capacity clamp begins to harden
$D_{\text{max}}^{\text{active}}$	8	Active disputes per ticket
$D_{\text{max}}^{\text{total}}$	32	Total accepted disputes per ticket
$B_{\text{max}}^{\text{queue}}$	2 MiB / ticket	Queued-claim metadata cap
Δ_{cr}	128 service opportunities	Challenge commit-reveal deadline
B_{overflow}	$64E_w$	Dedicated overflow-signal bond
$\Delta_{\text{challenge}}$	691,200 blocks	About 24 h at 8 BPS (informative)
I_w	256 ALGO-3 blocks	Target WinAnnounce interval (about 32 s)
Δ_{age}	65,536 ALGO-3 blocks	Minimum replica-lease age (about 2.3 h)
$N_{\text{ns}}^{\text{max}}$	4 / 65,536 blocks	Automatic suspension threshold
Δ_{setLead}	691,200 ALGO-3 blocks	Compute Set activation lead (about 24 h)
Δ_{stable}	4096 ALGO-3 blocks	Shadow-settlement stability depth (about 8.5 min)
H_{half}	$\geq 10\Delta_{\text{announce}}$	WinAnnounce-rate controller stability condition
S_{max}	1	Active Testnet-23 Compute Sets
$N_{\text{slots}}^{\text{max}}$	4096	Replica snapshot/selection memory cap
$B_{\text{weights}}^{\text{max}}$	Exact genesis-manifest size	Local weight-storage cap
$p_{\text{det}}^{\text{min}}$ / slash coefficients	Signed parameter package	Value-Profile rationality gate; disabled if absent
PALW ordering weight	0	Preserve the ALGO-3 fork choice
PALW transferable reward	0	Zero-value conformance
Real slash	Disabled	Shadow slash ledger only

$T_{\text{exec}}, \Delta_{\text{rs}}, \Delta_{\text{post}}, B_{\text{perf}}, B_{\text{corr}}, S_{\text{ns}}, S_{\text{proto}}, S_{\text{mis}}, V_{\text{epoch}}$, and fee coefficients are derived from Compute Set benchmarks and test-token policy. If a signed parameter package does not contain concrete values, the corresponding object class remains disabled. Fixed few-second constants and placeholders are forbidden.

16 Full-Node Validation Order

A full node checks length, version, network, ruleset, hashes, and budgets before expensive verification. Every PALW control transaction in one block is applied sequentially in canonical subnetwork and transaction-index order, making budget, sequence, capacity, and first-wins outcomes unique. The normative order is:

1. wire version, network, ruleset, ComputeVMID, and canonical length;
2. parent-branch budget epoch and PALW control-band state;
3. BatchCommit signature, authority sequence, anchor, fee, q , and one-shot state;
4. batch-specific lottery window, target reference, and T_{ticket} ;
5. WinAnnounce first-wins state, C_{lot} opening, winning trial, job leaf, job bytes, and escrows;
6. fixed roster root, verification window, R_{rep} , and R_{audit} ;
7. capacity token, canonical assignment, and performance-bond reservation;
8. replica commitments, primary opening, C_{exec} opening, NISA, replica witnesses, and projection equality;
9. Certificate signatures and correctness-lien reservation;
10. PendingTicket, challenge commit/reveal, bisection transition, queue caps, and verifier budget;
11. state-dependent maturity, automatic MatureTicket finalization, and nullifier; and
12. fees, shadow slashes, refunds, retention, and accumulator transition.

If a dependency is unavailable, a node places the object in pending state rather than guessing that it is valid. Missing and invalid are distinct. An online external service cannot complete consensus validity.

17 Launch Gates

Public Testnet-23 remains closed until all of the following hold.

- G1. Normative artifacts.** The Compute VM package, wire contract, positive/negative vectors, and ruleset hash are public, and independent runtime and node implementations agree. Production binaries contain no always-win target, mock Certificate, or legacy-wire fallback.
- G2. Determinism.** Benign mismatch is zero across CPU/GPU vendors, long context, MoE boundaries, restart, checkpoint restore, overflow, and malformed tokenizer input.
- G3. One-shot and reorganization.** Sequence, first-wins state, capacity, bonds, disputes, maturity, and nullifiers agree under randomized reorganization, restart, and clean IBD.
- G4. Attempt-side capacity.** BatchCommit rate, fee, live state, and control-band occupancy remain within epoch caps, and IBD catch-up exceeds chain generation.
- G5. Winner-side capacity.** p99.9 job bytes, PendingTicket adjunct size, verifier cost, burst propagation, and database growth remain within budgets.
- G6. Roster.** Independent credentials, free-slot concentration, utilization $\leq U_{\text{max}}$, no-shows, escalation, and lien freeze are measured for 30 days.
- G7. Dispute.** Honest and false claims, timeout, malformed moves, maximum active/total sets, and queue caps complete to one-step adjudication.
- G8. Control stability.** Under supply shock, replica outage, and capacity saturation, the target converges to accepted WinAnnounce rate without sustained oscillation.
- G9. Partition.** A 50/50 partition, reconnection, in-window reorganization, and late join converge naturally without manual arbitration.
- G10. Runtime tax.** Canonical-runtime throughput, checkpoint overhead, and padding ratio are published and remain within admission limits.
- G11. Watcher exercise.** Independent replay, challenge commit-reveal, overflow rejection, and bounty ownership are demonstrated on the live network.
- G12. Replica fault injection.** For every matrix row—no-show, equivocation, authenticated malformed body, opening mismatch, invalid initial/post-audit witness, wrong target/roster/assignment, and false Certificate—the live network demonstrates detection, one-shot freeze, suspension, shadow slash, bounty, and reorganization rollback; unauthenticated invalid evidence must never slash.
- G13. Bond rationality.** A Value Profile publishes $p_{\text{det}}^{\text{min}}$, maximum bribe, irrecoverable slash, and suspension cost and satisfies the replica-rationality inequality. Testnet-23 measures this gate but does not activate value.
- G14. Fail-closed profile.** Fork-choice weight, transferable reward, and real slashing are identically zero/disabled in every binary.

Passing these gates does not activate a Value or Ordering Profile. Those require a separate specification, external audit, monetary policy, GHOSTDAG safety analysis, and hard fork.

18 Limitations

1. NISA is a statistical audit, not a succinct proof.
2. Compute Set work pricing assumes the absence of undisclosed shortcuts. The protocol does not prove semantic usefulness or truthfulness.
3. Full nodes retain active weights; active-set count and total bytes are consensus-capped.
4. Full verification beyond the archival horizon requires archival providers.

5. Hash-lane majority withholding and targeted censorship are not eliminated, although PALW has no ordering weight in Testnet-23.
6. A replica Certificate is a pair of bonded attestations plus sampled witnesses, not a cryptographic proof of independent execution. Receipt of a correct trace from the primary is not objectively distinguishable from local replay; safety therefore relies on detecting and slashing false tickets and protocol faults.
7. Testnet-23 shadow bonds and zero-value tickets do not establish Value-Profile economic security. Rational honesty from substantial collateral is an explicit Value-Profile assumption and activation inequality, not a Testnet-23 conclusion.
8. Bounded, well-funded spam may reduce PALW ticket throughput but cannot become an ALGO-3 ordering failure.
9. Adding a Compute Set requires coordinated activation and may lag model obsolescence.

19 Conclusion

PALW v0.6.1 converts canonical LLM computation into compact hash trials and sends only winning executions through three-layer verification. Homogeneous BatchCommit and winner-only data availability bound attempt-side bandwidth and collateral lock. Separating lottery and result salts preserves immediate win verification while retaining blind replay. The operation accumulator commits computation that would otherwise disappear between checkpoints, and local weight retention keeps interval witnesses implementable.

B and C are treated as economically rational replicas with substantial bonds, but the protocol does not rely on goodwill. Complete field binding in CertificateBody, a future audit after signatures, objective timeouts, permissionless replica evidence, and one-shot fault state create multiple detection routes for fraudulent tickets, double signing, opening refusal, invalid witnesses, and false Certificates. Accepted evidence necessarily advances to collateral freeze, credential suspension, and slash settlement. A Certificate does not directly create a block or reward. A PendingTicket carries bounded disputes and becomes a MatureTicket only after every game resolves. In the Testnet-23 Safety Profile, the MatureTicket is a zero-value record and ALGO-3 alone provides ordering and liveness. The design therefore fails closed: uncalibrated useful-work economics, replica capture, witness defects, or controller instability cannot affect fork choice or real assets.

A future profile that grants PALW non-zero ordering power must not be enabled as a quiet extension of this document. It is a new protocol revision requiring renewed analysis of GHOSTDAG safety, external work recovery, monetary policy, and implementation evidence.

A Response to Third-Audit Findings

ID	Prior finding	Normative v0.6.1 correction
C-01	Dual block-before-dispute and pre-block flows	One flow: PendingTicket $\rightarrow$ dispute $\rightarrow$ MatureTicket. Testnet-23 creates no PALW ordering block.
C-02	Inconsistent Δ_{exp} and challenge lifetime	Separate Δ_{cert} , $\Delta_{\text{challenge}}$, and state-dependent maturity.
C-03	Revealed lottery salt made the projection commitment testable	Separate $C_{\text{lot}}/C_{\text{exec}}$ and $\rho_{\text{lot}}/\rho_{\text{res}}$.
C-04	All-attempt job bytes and escrow dominated resource use	Homogeneous batching, winner-only job bytes, q -proportional fee, and escrow only at WinAnnounce.
C-05	Long-lived reservations required an impractically large roster	B_{perf} lasts to Certificate; B_{corr} lasts to maturity, with utilization hard caps.
C-06	Weight-tile witnesses forced over-fine checkpoints	Full nodes retain active weights; witnesses contain state, paths, and opcode slices.
H-01	Equivocation proof could not execute	Testnet-23 has no PALW block authority or ordering ticket; the mechanism is removed from scope.
H-02	Shared endpoint redrew all commitments together	Power-of-two w_{spread} offset derived from carrier hash and BatchID.
H-03	Announcement timing selected the roster snapshot	Roster root is frozen at lottery-window end.
H-04	Capacity skipping amplified attacker free share	Global utilization cap, credential cap, free-share bound, and canonical assignment.
H-05	Fixed 32-block replay deadline	Deadline proportional to measured $T_{\text{exec}}(s, \sigma)$.
H-06	Congestion caused honest slashing	Dedicated control band, service-clock pause, and separate treatment of targeted censorship.
H-07	Dead computation was invisible at checkpoints	Rolling accumulator for every semantic operation, padding cap, and shortcut analysis.
H-08	Dispute delay destabilized ASERT	WinAnnounce-rate controller, capacity clamp, and $H_{\text{half}} \geq 10\Delta_{\text{announce}}$.
H-09	Short use window enabled targeted DoS	Deterministic automatic finalization; no authority signature or use window.

ID	Prior finding	Normative v0.6.1 correction
H-10	$\chi = 1$ underpriced ordering weight	PALW ordering weight fixed to zero in Testnet-23; non-zero profiles are separate specifications.

B Additional Norms for B/C Detection and Slashing

This revision turns the premise that substantially bonded B and C behave rationally into verifiable protocol conditions: (i) the replica-rationality inequality, (ii) evidence completeness through valid outer signatures or objective timeout, (iii) a Certificate-Body binding target, roster, assignment, and execution evidence, (iv) a future audit after B/C signatures, (v) OffenseID and one-shot FaultState, (vi) deterministic freeze, suspension, and slash for every fault-matrix row, and (vii) an attribution rule that never slashes for an unauthenticated invalid signature. These rules do not claim that every false execution is detected with certainty. Detection may depend on sampled audits and watchers, but the normative protocol contains no discretionary path that accepts objective evidence and declines to slash.

C Normative Artifact Checklist

A consensus release binds the following roots into one RulesetID:

$$\text{RulesetID} = H(r_{\text{wire}} \parallel r_{\text{vm}} \parallel r_{\text{set}} \parallel r_{\text{params}} \parallel r_{\text{vectors}}). \quad (64)$$

Runtime and node independently rederive BatchID, windows, target, trial, samples, assignments, Certificate, dispute, maturity, and settlement from the same input. Regenerating golden vectors is a ruleset change and requires a coordinated release plus a fresh genesis or explicit activation fence.

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